order47 RESULTS: exact certification
PREREG sha256 72d93d589f678bad16706dd06d0fdae0255c3ad1b4eaee21425ecdde49158e39 (hashed before the run).
Method: exact integer arithmetic in Z[ζ_9800]. Products are formed by exponent addition mod 9800. The zero test uses the Z-basis ζ^r·ζ^{140i} (r < 140, i < 24), via Φ_9800(x) = Φ_70(x^140), with exact reduction mod Φ_70. No floating point in the verdicts.
identity	nonzero entries of LHS − RHS (of 105^2)	verdict
control: $M^2 = 35 \cdot P$	0	PASS (machinery validated)
control: M·M^† = 34·I (wrong on purpose)	105	rejected, as required
X1: conj(M)·Π·M = 35·Π	0	**PASS (exact)**
X2: M·M^† = 35·I	0	**PASS (exact)**
Run time ~2 s.
Consequence (as fixed in PREREG)
With fact (i) (Y(−x) = conj(ΠY(x)) for real-weight sums; proved), the order45 identity is now **Category A, proved**:
R_Y(1/x) = −x^{−1}·conj(Π(kM)^{−1}R_Y(x)) and ‖R_Y(1/x)‖ = ‖R_Y(x)‖/|x|.
v11: YES
papers/RTA_FOUNDATION_BOUND/RTA_FOUNDATION_BOUND_v11.pdf, 135 pages (new files: main_v11, front_v11, partIII_inv_v11, appD_v11, NOTES_v11.md; v10 untouched).
New: Proposition (exact, computer-assisted) conj(M)ΠM = 35Π and MM^† = 35I; Theorem (x ↔ 1/x symmetry for real-weight sums); Category B remark; App. D item.

order47 NOTES
- The theorem is stated for all real-weight sums. The numerical evidence so far is for C2b only, so order48 checks the prediction on C2a, C11, A2_corner (CO) and the lines using stored arrays (reciprocal pairs inside XS).
- Naming: RG* only, no labels L1–L7; "36" only as dim V. MAPS_CANDIDATES untouched; no Resonant objects; no bridge claim; CORE_FREEZE; no push or sends; new files only.